Intelligent Decision Support System for Sri Lankan Tea Estate Management

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DECLARATION OF THE CANDIDATE & SUPERVISOR

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ABSTRACT

Tea grading is a critical process in Sri Lankan tea factories because the assigned grade directly influences product quality, pricing, production planning, and market confidence. Traditionally, tea sorting and grading are largely performed through manual visual inspection and sieve-based separation, relying heavily on the experience and judgement of factory operators. Although this approach is widely practiced, it can be affected by human subjectivity, operator fatigue, variations between tea batches, and the absence of consistent digital decision-support mechanisms. These challenges can lead to variations in grading outcomes and limit the availability of reliable data for quality monitoring and production management.

This research proposes an AI-based Tea Sorting and Grading Automation module as part of an intelligent decision support system for Sri Lankan tea estate management. The system applies computer vision and Convolutional Neural Networks (CNNs) to automatically classify processed tea leaves into nine grades: BM, BOP, BT, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust. The classification is performed using visual characteristics captured from tea samples, including leaf size, shape, texture, colour, and degree of breakage. A tea-grade image dataset will be created using samples collected under controlled imaging conditions, followed by image preprocessing, dataset preparation, model training, and performance evaluation.

The trained CNN model will be evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis, to determine its effectiveness in distinguishing between the different tea grades. In addition to individual image classification, the system will provide confidence-based predictions, batch-wise grade summaries, and digital production reports. These features are intended to support quality control officers and factory managers by providing faster, more consistent, and data-driven grading information.

The research aims to demonstrate the practical applicability of AI-based tea grading within Sri Lankan factory environments while reducing dependence on subjective manual assessment. By integrating automated image classification with digital reporting and decision support, the proposed solution is expected to improve grading consistency, support quality monitoring, reduce operator workload, and enhance production-level decision making. Furthermore, the system has potential for future commercial deployment as an affordable AI-assisted quality control solution for small, medium, and large-scale tea manufacturing facilities, contributing to the digital transformation and competitiveness of Sri Lanka's tea industry.

Keywords: Tea Grading, Computer Vision, Convolutional Neural Network, Image Classification, Quality Control

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligent
BOP	Broken Orange Pekoe
CNN	Convolutional Neural Networks
DSS	Decision Support System
FBOP	Flower Broken Orange Pekoe
ML	Machine Learning
SDG	Sustainable Development Goal
SLTB	Sri Lanka Tea Board
UI	User Interface

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1. INTRODUCTION

The tea industry remains one of Sri Lanka's most important agricultural and export sectors. Ceylon Tea has a strong international reputation for quality, and the industry makes a significant contribution to the country's export earnings. The quality and consistency of made tea influence auction prices, buyer confidence, brand reputation, and the long-term competitiveness of the industry. Within the tea manufacturing process, sorting and grading are therefore important quality-control activities that determine how tea products are classified, evaluated, marketed, and sold.

In practice, tea sorting and grading are commonly supported by mechanical separation, sieve-based processes, and human visual assessment. Trained factory personnel inspect tea particles and identify grades based on characteristics such as particle size, appearance, texture, color, and degree of breakage. However, manual assessment can be influenced by human subjectivity, operator fatigue, differences in experience, and variations between tea batches. These factors can lead to inconsistencies in grading and may affect subsequent production and quality-control decisions.

Manual grading also creates operational challenges because the process can be time-consuming and difficult to standardize across different operators, shifts, and factory environments. In addition, when grading information is not digitally recorded, factories have limited opportunities to analyze historical grading patterns, uncertain classifications, and recurring quality variations. Consequently, management decisions may rely heavily on individual observations rather than structured and accessible production data.

Recent developments in Artificial Intelligence (AI), particularly computer vision and deep learning, provide opportunities to improve image-based quality assessment in agricultural and manufacturing environments. Convolutional Neural Networks (CNNs) are particularly suitable for image classification because they can automatically learn relevant visual features from image data. This makes CNN-based

approaches applicable to tea grading, where visual characteristics of processed tea particles play an important role in distinguishing different grades.

This research proposes an AI-Based Tea Sorting and Grading Automation System that uses tea sample images to classify processed tea into nine grades: BM, BT, BOP, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust. The images will be collected under controlled, factory-related conditions and used to develop and evaluate a CNN-based classification model. The system is intended to function as an intelligent decision-support tool rather than completely replace human expertise. It will provide automated classification, confidence-based predictions, batch-level grade summaries, and digital production reports to support quality-control officers and factory managers.

Statistical Context - Sri Lanka remains a major global tea-producing and exporting country. According to Sri Lanka Tea Board statistics, tea exports in 2022 totaled approximately 250.17 million kg, with total tea export value reaching approximately Rs. 411.05 billion. The Tea Board also reported an estimated tea production of approximately 290 million kg and expected export earnings of around USD 1.4 billion for 2023. These figures demonstrate the economic importance of maintaining consistent quality throughout the tea production and export process.

SDG Alignment - This research aligns with United Nations Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) by applying Artificial Intelligence and computer vision to modernize quality-control processes within Sri Lanka's tea manufacturing sector. It also supports SDG 8 (Decent Work and Economic Growth) by promoting productivity, technological innovation, and sustainable economic development within an important national industry. Furthermore, improved digital quality monitoring can contribute to SDG 12 (Responsible Consumption and Production) by supporting more consistent production processes and reducing quality-related inefficiencies.

Research Cluster Alignment - This research falls within the Intelligent Systems and Data Science research area, with a specific focus on applied machine learning, computer vision, image classification, and intelligent decision-support systems. It is

also relevant to Digital Transformation and Smart Agriculture, as it applies AI technology to a traditional Sri Lankan agro-industrial domain and demonstrates how digital technologies can support quality monitoring and operational decision-making.

1.1 Background and Literature Review

Tea grading is an important quality assessment activity within the tea manufacturing process, as the characteristics of processed tea determine how the final product is categorized and evaluated. Conventional tea grading generally considers visual characteristics such as particle size, appearance, colour, texture, and uniformity. In Sri Lankan tea manufacturing, experienced workers and tea professionals have traditionally played an important role in evaluating these characteristics. However, the grading process can be labour-intensive and time-consuming, while the dependence on human observation can introduce subjectivity into the classification process. A recent study on Sri Lankan leafy tea grades emphasized that tea grading relies heavily on the knowledge and experience of tea workers and proposed deep learning as an alternative for automated classification [1]. These limitations have encouraged researchers to investigate computer vision and machine learning techniques for improving the consistency and efficiency of tea quality assessment.

The application of computer vision to agricultural products has expanded significantly with the development of machine learning and deep learning techniques. Image-based approaches allow visual characteristics to be captured digitally and analyzed using computational models rather than relying entirely on manual observation. In tea-related research, early classification approaches have demonstrated that visual information can be used to distinguish different tea categories. Li et al. developed a five-layer customized Convolutional Neural Network (CNN) for classifying three tea categories and reported an accuracy of 97.96%, demonstrating the potential of CNNs for tea image classification [2]. Although this study focused on broad tea categories rather than processed Sri Lankan factory grades, it established that deep learning can identify meaningful visual patterns from tea images.

More recent research has focused specifically on tea quality and grade classification. Amarabandu and Rupasinghe investigated the classification of five Sri Lankan leafy tea grades, namely BOP1, OP1, OPA, Pekoe, and Chunmees, using deep learning [1]. The researchers created a dataset containing 2,500 tea images and evaluated CNN, MobileNetV2, and VGG16 models. Their findings demonstrated that deep learning models can effectively distinguish visually similar tea grades, with VGG16 achieving approximately 99% classification accuracy. The study is particularly relevant to the Sri Lankan context because it demonstrates the feasibility of using image-based deep learning for local tea-grade classification. However, the study concentrated on five leafy tea grades and did not address a broader set of processed factory grades containing categories such as dust and fannings.

Other studies have further demonstrated the effectiveness of deep learning for tea-related image analysis, although their objectives differ from direct tea-grade classification. For example, research on tea leaf disease classification has used deep learning models to identify different visual conditions in tea leaves. A recent study developed a curated tea-leaf dataset and applied a lightweight deep learning framework using standardized image preprocessing and augmentation techniques, achieving a classification accuracy of 96.80% [3]. Similarly, research on identifying the geographical origin of tea leaves has demonstrated that lightweight deep learning models can extract subtle visual characteristics from tea images. Pei et al. developed a lightweight CNN-based approach using images collected from different tea-growing regions and achieved an average overall accuracy of approximately 92% [4]. These studies indicate that deep learning can recognize fine-grained visual differences in tea-related images; however, they primarily focus on leaf disease and geographical origin rather than the classification of processed tea grades used in factory production.

The success of CNNs in tea image classification is also supported by their broader application in agricultural image analysis. CNNs automatically learn hierarchical visual features from images, allowing models to identify patterns such as shapes, textures, edges, and colour variations without requiring all features to be manually designed. The development of architectures such as VGG, ResNet, MobileNet, and

EfficientNet has further improved the flexibility of deep learning for image classification tasks [5]–[8]. VGG networks demonstrated the effectiveness of deeper convolutional architectures for visual recognition, while ResNet introduced residual connections that enabled the training of substantially deeper networks [5], [6]. MobileNetV2 subsequently provided a computationally efficient architecture using inverted residuals and linear bottlenecks, making it suitable for applications where computational resources are limited [7]. EfficientNet further explored the balance between model depth, width, and image resolution to achieve improved performance with comparatively efficient architectures [8].

Transfer learning has also become an important approach for agricultural image classification, particularly when large domain-specific datasets are difficult to obtain. Instead of training a CNN entirely from the beginning, a model pretrained on a large image dataset can be adapted to a specific agricultural classification task. A recent review of transfer learning in agriculture reported that pretrained deep learning models can provide strong performance in agricultural applications where labelled datasets are limited [9]. This approach is particularly relevant to tea grading because obtaining a large number of accurately labelled tea images from factory environments can require substantial time and involvement from experienced personnel. Fine-tuning pretrained architectures such as VGG16, ResNet, MobileNet, or EfficientNet can therefore provide an effective strategy for learning tea-specific visual characteristics while reducing the amount of training data required compared with training a deep network from scratch.

The quality and diversity of the image dataset are also important factors influencing the performance of deep learning models. Agricultural images can vary because of differences in lighting, camera position, background, object orientation, and image quality. These variations can affect the ability of a model to generalize to unseen samples. Research on tea image datasets has demonstrated the importance of standardized preprocessing and image preparation. For example, recent tea-leaf research applied image resizing, normalization, background processing, and augmentation before model training [3]. Similarly, another tea image dataset study

used manually labelled images collected under field conditions and applied preprocessing to improve consistency before evaluating a ResNet50 model [10]. These findings suggest that careful dataset construction and preprocessing are essential when developing an automated tea classification system.

Data augmentation provides another approach for improving model generalization when the available dataset is limited. Augmentation techniques can create variations of existing training images through transformations such as rotation, flipping, brightness adjustment, and other controlled modifications. Such techniques can help models become less dependent on specific image orientations or lighting conditions. Shorten and Khoshgoftaar reviewed image augmentation techniques for deep learning and highlighted their usefulness in improving model generalization and reducing overfitting [11]. Recent tea image classification research has similarly used rotation, flipping, and brightness adjustments to increase the robustness of deep learning models [3]. Therefore, suitable augmentation strategies can be particularly useful for tea-grade datasets where collecting large numbers of independent samples for every grade may be difficult.

Another important consideration is the selection of appropriate evaluation measures for multi-class image classification. Overall accuracy provides a general indication of how many samples are correctly classified, but accuracy alone may not adequately represent model performance when different classes have unequal sample distributions. Precision measures the proportion of predicted samples that actually belong to a particular class, while recall measures the proportion of actual samples that are correctly identified. The F1-score combines precision and recall and provides a useful measure for evaluating classification performance across individual classes [12]. Confusion matrices can additionally identify specific grades that are frequently confused with one another. These measures are important for tea grading because visually similar grades may be more difficult to distinguish than visually distinct categories. Therefore, evaluating the model using accuracy together with precision, recall, F1-score, and a confusion matrix can provide a more comprehensive assessment of classification performance.

The use of confidence information can further improve the practical value of an automated classification system. Deep learning classifiers generally produce probability values for the predicted classes, which can be interpreted as an indication of the model's confidence in its prediction. However, research has shown that the confidence values produced by modern neural networks are not always perfectly calibrated and may sometimes be overconfident [13]. For a tea grading application, confidence-based predictions can therefore be used as decision-support information rather than as an absolute replacement for expert judgment. Samples associated with low confidence can be identified for additional human inspection, while high-confidence predictions can be incorporated into digital batch-level reports. This approach can support a human-in-the-loop quality-control process in which artificial intelligence assists factory personnel rather than completely replacing their expertise.

Although computer vision and deep learning have shown promising results in tea-related classification tasks, important limitations remain in the existing literature. Many studies focus on fresh tea leaves, disease detection, geographical origin, or a limited number of tea categories [2]–[4], [10]. The recent Sri Lankan study by Amarabandu and Rupasinghe provides strong evidence for automated classification of leafy tea grades, but its classification scheme is limited to five grades and does not cover a broader set of processed tea categories used in factory-level production [1]. In particular, there is limited research addressing multiple processed tea grades that include both larger particle grades and finer categories such as dust and fannings within a single automated classification framework. This creates an opportunity to investigate computer vision and CNN-based methods for a more comprehensive processed-tea grading problem.

Furthermore, existing research has primarily concentrated on achieving classification accuracy rather than integrating classification results into a practical quality-control workflow. An automated grading system can provide greater value when predictions are accompanied by confidence information, batch-level summaries, and digitally recorded production reports. Such features can allow quality-control personnel and factory managers to review classification outcomes, identify uncertain samples, and

analyze grading patterns across production batches. Therefore, the development of an intelligent tea grading system should consider not only the performance of the underlying CNN model but also how its outputs can support real-world quality-control decisions.

Based on the reviewed literature, deep learning and computer vision provide a promising foundation for automating tea image classification. Previous studies demonstrate that CNN-based approaches can achieve high classification performance in different tea-related applications, while transfer learning and data augmentation can help address the challenges associated with limited agricultural datasets [1]– [4], [9], [11]. Nevertheless, there remains a need for research specifically focused on automated classification of multiple processed Sri Lankan tea grades within a single system. This research addresses this gap by developing an AI-based tea sorting and grading automation system capable of classifying BM, BOP, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust from tea sample images. By combining image-based CNN classification with confidence-aware predictions and digital production reporting, the system aims to extend existing research from experimental tea image classification toward a more practical decision-support solution for tea-factory quality control.

1.2 Research Gap

Although recent studies have demonstrated the potential of deep learning and computer vision for tea-related image classification, significant gaps remain in the automated grading of processed tea in the Sri Lankan tea industry. Existing research has mainly focused on tea category classification, fresh tea leaf disease identification, geographical-origin classification, or a limited number of leafy tea grades [1]–[4]. For example, Amarabandu and Rupasinghe [1] demonstrated that deep learning can effectively classify five Sri Lankan leafy tea grades, while Li et al. [2] showed the feasibility of CNN-based classification for different tea categories. However, these studies do not specifically address the classification of multiple processed tea grades commonly handled within Sri Lankan tea factories.

A further gap is the limited research on automated image-based classification of a broader range of processed tea grades within a single system. This research focuses specifically on nine processed tea grades: BM, BOP, BT, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust. Existing studies generally investigate a smaller number of classes or address different tea-related classification problems [1]–[4]. Therefore, there is a need for a dedicated classification approach that can distinguish multiple processed tea grades based on their visual characteristics.

Another limitation identified in previous research is the gap between experimental image classification and practical quality-control applications. While existing studies report classification performance using metrics such as accuracy, precision, recall, and F1-score, fewer studies consider how automated predictions can be integrated into a practical tea-factory workflow. In particular, there is limited attention to confidence-aware predictions, batch-level grade summaries, and digital production reporting that can support quality-control personnel and factory management.

Dataset characteristics also represent an important research gap. Deep learning performance is strongly influenced by the quality, diversity, and preparation of training images. Previous tea-related studies have used datasets collected under different conditions, including fresh leaves, field environments, and specific experimental

settings [3], [4], [10]. However, there is limited research using image datasets specifically designed for the visual classification of multiple processed Sri Lankan tea grades under controlled, factory-related conditions.

Therefore, the main research gap identified from the literature is the lack of an integrated AI-based system specifically designed to classify multiple processed Sri Lankan tea grades while providing practical decision-support features for tea-factory quality control. This research addresses this gap by developing a CNN-based tea sorting and grading automation system capable of classifying BM, BOP, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust from tea sample images. In addition to automated classification, the system incorporates confidence-based predictions, batch-level grade summaries, and digital production reports, thereby extending existing research from standalone tea image classification toward a more practical and industry-oriented quality-control solution.

Table 1.1 Comparative Analysis of Existing Studies

Study	**Main Focus**	**Tea Type / Classes**	**AI / ML Approach**	**Main Limitation / Gap**
Li et al. [2]	Tea category classification	3 tea categories	Customized CNN	Limited number of categories; not focused on processed Sri Lankan tea grades
Amarabandu & Rupasinghe [1]	Sri Lankan tea-grade classification	5 leafy tea grades	CNN, MobileNetV2, VGG16	Focuses on leafy tea grades rather than multiple processed factory grades
Chowdhury et al. [3]	Tea leaf disease classification	4 disease conditions	Deep learning	Focuses on disease detection, not tea-grade classification
Pei et al. [4]	Tea geographical-origin identification	Tea leaves from different regions	Lightweight deep learning	Focuses on geographical origin rather than processed tea grading

Partho et al. [10]	Tea leaf condition classification	4 leaf conditions	ResNet50	Uses fresh tea leaves and does not address processed tea grades
Proposed System	**Automated processed tea sorting and grading**	**8 processed tea grades: BM, BOP, BP, Broken Tea, Dust, Fanning 2, PF, PW Dust**	**CNN-based image classification**	**Addresses multi-grade processed tea classification with confidence-based predictions, batch summaries, and digital reports**

1.3 Research Problem

The Sri Lankan tea industry plays a significant role in the country's economy, making consistent tea quality assessment essential for maintaining product quality, market competitiveness, and export value. However, the sorting and grading of processed tea still rely substantially on mechanical separation methods and human visual assessment. These approaches can be affected by differences in operator experience, subjective judgment, and variations in the visual characteristics of tea samples, potentially leading to inconsistencies in grading decisions.

Although recent studies have demonstrated the effectiveness of Convolutional Neural Networks (CNNs) and other deep learning techniques in tea image classification, existing research has primarily focused on fresh tea leaves, tea leaf diseases, geographical origin identification, or a limited number of leafy tea grades. Consequently, there is a lack of research specifically addressing the automated image-based classification of multiple processed Sri Lankan tea grades, including BM, BOP, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust, within a single classification framework.

Furthermore, automated tea classification alone may not fully address the practical requirements of tea-factory quality control. There is a need for a system that not only classifies tea samples based on their visual characteristics but also provides confidence-based predictions, batch-level grade summaries, and digital production reports to support quality-control personnel in reviewing classification results and making informed decisions.

Therefore, the main research problem is the lack of an integrated, AI-based system specifically designed to classify multiple processed Sri Lankan tea grades from images while providing reliable classification results and practical digital decision-support features for tea-factory quality control. This research aims to address this problem by developing and evaluating a CNN-based tea sorting and grading automation system for nine processed tea grades, supported by confidence-aware predictions and digital reporting capabilities.

1.4 Research Objectives

1.4.1 Main objective

The main objective of this research is to develop and evaluate an Artificial Intelligence (AI)-based tea sorting and grading automation system using Convolutional Neural Networks (CNNs) to classify nine processed Sri Lankan tea grades from digital images. The system aims to improve the consistency and efficiency of tea-grade classification by automating visual assessment and integrating confidence-based predictions, batch-level grade summaries, and digital production reporting to support quality-control personnel and factory management in making informed grading decisions.

1.4.2 Specific objective

- **To collect, label, and prepare a dataset of processed Sri Lankan tea images representing nine tea grades.**

This objective focuses on developing a suitable image dataset for training and evaluating the classification model. Tea sample images representing BM, BT, BOP, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust will be collected under controlled, factory-related conditions. The collected images will be labelled according to their respective tea grades and organized into appropriate classes. Image preprocessing techniques, including resizing, normalization, and suitable data augmentation, will be applied where necessary to improve dataset consistency and support model training. The dataset will also be examined for class imbalance, image quality, and variations in visual characteristics to ensure its suitability for multi-class classification.

- **To develop and train a CNN-based image classification model for identifying the nine processed tea grades.**

  This objective involves designing and implementing a deep learning model capable of identifying visual differences among the nine selected tea grades. The model will learn relevant image features, including particle size, shape, texture, colour, and distribution patterns, to distinguish between visually similar categories. Appropriate CNN architectures and, where suitable, transfer learning techniques will be investigated to identify a model that provides an effective balance between classification performance and computational requirements. The prepared dataset will be divided into training, validation, and testing subsets to support model development, parameter tuning, and independent performance evaluation.

- **To evaluate the classification performance and reliability of the developed CNN model using appropriate evaluation metrics.**

  This objective aims to assess how accurately and consistently the trained model identifies the nine processed tea grades. Model performance will be evaluated using accuracy, precision, recall, and F1-score, together with a confusion matrix to identify correctly and incorrectly classified samples. Per-class performance will also be examined to determine whether particular tea grades

are more difficult to distinguish because of similarities in their visual characteristics. Where appropriate, macro-averaged metrics will be used to assess performance across all nine classes. The evaluation results will be used to identify the model's strengths, limitations, and suitability for supporting tea-factory quality-control activities.

- **To integrate the trained CNN model into a web-based system that provides automated tea-grade predictions and confidence scores.**

  This objective focuses on incorporating the trained classification model into an accessible web-based application. The system will allow users to upload processed tea sample images and obtain predicted tea grades without relying entirely on manual visual classification. For each prediction, the system will display the identified tea grade and the associated model confidence score. Appropriate confidence thresholds may be established to flag uncertain predictions for additional inspection by trained personnel. The system will be designed to support a human-in-the-loop approach in which automated predictions assist quality-control decisions while preserving the role of human expertise in reviewing uncertain cases.

- **To implement batch-level grade summaries and digital production reports to support tea-factory quality-control and decision-making processes.**

  This objective involves developing digital reporting features that organize and summarize classification results for multiple tea samples within a batch. The system will present the distribution of predicted grades and relevant classification information in a structured format. It will also generate digital production reports that allow users to review, record, and retain grading results for future reference. These features are intended to improve the accessibility of grading information, support the identification of recurring classification patterns, and reduce reliance on unstructured manual records. The practical usefulness of the reporting functionality will be assessed in relation to the intended quality-control workflow.

2. METHODOLOGY

2.1 Methodology

This research adopted a system-development approach to implement an AI-based tea sorting and grading application for classifying processed tea samples using digital images. The methodology focused on developing a web-based application, integrating a remotely hosted machine learning classification service, managing prediction records, and providing digital reporting facilities to support tea-grade assessment.

The developed application, TeaGuard AI, follows a microservice architecture consisting of a React-based frontend, an API Gateway, an Authentication Service, and a Model Service. The Model Service communicates with an externally hosted classification model through a Gradio-compatible API, while MongoDB Atlas provides persistent storage for user information and classification records. The application supports the classification of nine tea grades: BM, BT, BOP, BP, BROKEN_TEA, DUST, FANNING_2, PF, and PW_DUST.

The methodology describes the implemented application and its integration with the remote classification service. Since the original model-training scripts, complete dataset documentation, and model weights were not available in the supplied source code, the model-training configuration and experimental performance are treated separately from the verified application implementation.

2.1.1 Research Approach

The research followed an applied system-development approach, combining machine learning integration, web application development, database management, and digital reporting. The primary objective was to implement a system that accepts tea sample images, obtains automated classification results from a remotely hosted model, and presents the results through a web interface.

The application was developed to support whole-image classification rather than individual tea-particle detection. Each uploaded image is submitted to the classification service, which returns a predicted tea grade and, where available, confidence information for the supported classes.

The implementation was organized into the following stages:

1. Identification of the functional requirements for tea-grade classification.
2. Development of the web-based user interface.
3. Implementation of the API Gateway and authentication mechanisms.
4. Development of the Model Service and integration with the external classification endpoint.
5. Integration of MongoDB Atlas for persistent data storage.
6. Implementation of classification history and digital reporting functions.
7. Verification of the application workflow and its component interactions.

2.1.2 System requirements

The system requirements define the main functions and quality characteristics required for the TeaGuard AI system to perform its intended tea-grade classification and supporting activities. The requirements were identified based on the objectives of the research and the practical requirements of the tea-grade classification process. They are divided into functional requirements, which describe what the system should do, and non-functional requirements, which describe the expected quality and operational characteristics of the system.

Functional Requirements

The TeaGuard AI system shall provide user authentication and authorization to ensure that only authorized users can access protected system functions. The system shall allow users to upload tea sample images for classification and validate the uploaded files according to the supported image format and maximum file size. The system shall

send the uploaded image to the classification service and obtain the predicted tea grade from the machine learning model.

The system shall support classification of the nine processed tea grades: BM, BOP, BT, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust. After classification, the system shall display the predicted tea grade and available confidence information to the user. The system shall store classification results and relevant image metadata in the database to support future reference. Users shall be able to access previously stored classification records through the classification history interface.

The system shall also provide reporting functionality that allows users to export relevant classification information in PDF and CSV formats. In addition, the system shall provide communication between the frontend, backend services, external classification service, and database to complete the classification workflow.

Non-Functional Requirements

The TeaGuard AI system shall provide reliable and consistent communication between its frontend, backend services, classification service, and database. The system should respond to classification requests within a reasonable time under normal operating conditions to provide a practical user experience.

Security is an important requirement of the system. User authentication shall be implemented using access-token-based authorization, while uploaded files shall be validated before being processed. The system should also protect stored classification records and restrict access to authorized users.

The system shall be maintainable through a modular architecture in which the frontend, API Gateway, Model Service, authentication service, and database can be managed independently. The system should also be scalable so that additional tea grades, users, or system functionality can be incorporated in future development.

The application shall provide a user-friendly interface that allows users to upload images, view classification results, access classification history, and generate reports

without requiring specialized technical knowledge. The system shall also maintain data integrity by storing classification records with relevant metadata and timestamps.

Finally, the system should be compatible with common modern web browsers and operate effectively within the supported deployment environment. The use of containerization and web-based services should support consistent deployment and easier maintenance across different environments.

2.1.3 Data collection

Data collection was a critical component of the research and was conducted in a structured and ethically responsible manner. The dataset used for the tea-grade classification research was collected from the tea factories with the assistance of quality-control officers. Their involvement was important in identifying and verifying the different processed tea grades based on the grading practices followed within the tea-manufacturing environment. The data collection process focused on obtaining representative tea samples belonging to nine predefined categories: BM, BT, BOP, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust.

The quality-control officers supported the identification and labelling of the collected tea samples according to their respective grades. Their knowledge and practical experience helped establish the reference labels required for developing and accessing an automated tea-grade classification system. Using samples obtained from the tea factories also helped align the dataset.

Digital images of the tea samples were collected to represent the visual characteristics of the nine grade categories. These images served as the input data for the image-based classification workflow. The collected dataset was intended to capture visual differences among the grades, including variations in particle size, shape, texture, color, and overall appearance. Such characteristics are relevant to distinguishing processed tea grades through image classification.

The involvement of tea-factories quality-control officers helped ensure that the grade labels were informed by practical grading knowledge rather than assigned solely through visual assumptions.

The data was used solely for research and model training purposes, ensuring compliance with ethical standards in educational research.

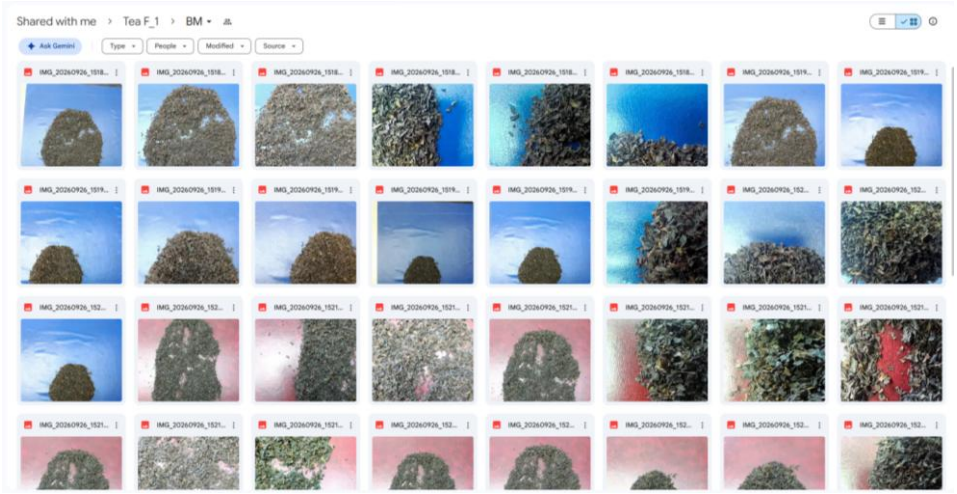


Figure 2:1 Sample data of BM data collection

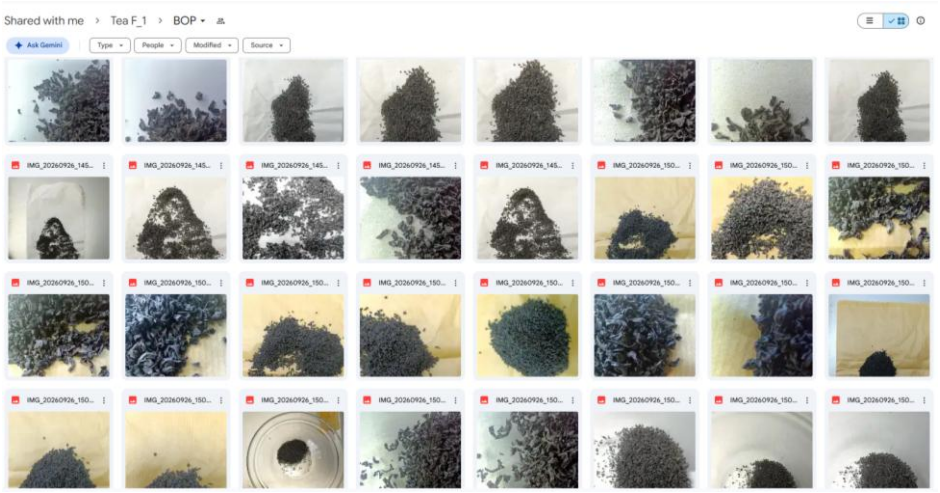


Figure 2:2 Sample data of BOP data collection

2.1.4 Tea Grade Classification Model Development

The tea-grade classification model was developed as a whole-image image-classification model to automatically identify the processed tea grade represented in an input image. Unlike object-detection approaches, the model does not detect, count, or draw bounding boxes around individual tea particles. Instead, each input image is assigned a single tea-grade label. The classification model was developed using a pretrained ConvNeXt V2 Tiny convolutional neural network (CNN), followed by transfer learning and fine-tuning for the tea-grade classification task.

Model Architecture

The selected model architecture was ConvNeXt V2 Tiny, using the model identifier convnextv2_tiny.fcmae_ft_in22k_in1k. The model was initialized using pretrained weights and subsequently adapted to the tea-grade classification task through transfer learning and fine-tuning. This approach allowed the pretrained visual features of the network to be adapted to the visual characteristics of processed tea samples.

The model performs image-level classification, where one tea-grade label is predicted for the complete input image. The trained classifier contains nine output classes: BM, BP, BT, Bop, Broken, Dust, Faning2, PF, and PWdust. The model development dataset contained 1,217 usable images.

Dataset Preparation and Splitting

The collected tea images were organized according to their corresponding tea-grade labels. The dataset was divided into training, validation, and testing subsets using an 80%/10%/10% split. This resulted in 973 images for training, 122 images for validation, and 122 images for testing. The training set was used to optimize the model parameters, while the validation set was used to monitor model performance during development. The test set was reserved for evaluating the final classification performance.

The image-level split represents the reported experimental evaluation; therefore, if multiple images originated from the same physical tea sample, the resulting test performance may be optimistic. Independent sample-level testing would provide stronger evidence of the model's performance on unseen physical tea samples.

Image Preprocessing

Before being provided to the classification model, images were prepared according to the model's required input configuration. The input image size was set to 224 × 224 pixels. The preprocessing configuration used normalization with a mean of (0.485, 0.456, 0.406) and a standard deviation of (0.229, 0.224, 0.225). The model preprocessing configuration also specified center cropping with a crop percentage of 0.875.

Training augmentation was included as part of the previous training workflow; however, the exact transformation settings for operations such as rotation, horizontal or vertical flipping, and brightness modification could not be confirmed from the available run information. Therefore, specific augmentation values are not stated as fixed experimental parameters.

During dataset validation, one oversized image belonging to the PWdust category was skipped. The remaining valid images formed the usable dataset used for model development.

Training Configuration

The model was trained for a total of 80 epochs. The training schedule consisted of 20 epochs of classification-head training followed by 60 epochs of full-model fine-tuning. A batch size of 32 was used, and the model received images at an input resolution of 224 × 224 pixels.

Transfer learning was used to adapt the pretrained ConvNeXt V2 Tiny model to the tea-grade classification problem. The training process was conducted using PyTorch and the timm model library.

Training Environment

The model development and training process was conducted using Python in Google Colab. PyTorch was used as the deep learning framework, while the timm library was used to access and configure the ConvNeXt V2 Tiny architecture. Training was performed using an NVIDIA L4 GPU with CUDA support. Google Drive was used for data and model-related storage during the development process.

Model Evaluation

The trained model was evaluated using the independent test subset containing 122 images. Several classification metrics were used to assess its performance, including accuracy, macro-averaged precision, macro-averaged recall, and macro-averaged F1-score.

The model achieved a test accuracy of 94.26%. The macro-averaged precision was 95.99%, while the macro-averaged recall was 94.63%. The macro-averaged F1-score was 94.72%. The best validation macro F1-score recorded during model development was 96.44%.

The reported test loss was 0.6431, while the final training loss and final validation loss were 0.5437 and 0.5982, respectively. These results indicate that the model achieved strong classification performance on the reported test split. However, the results should be interpreted in the context of the image-level dataset split, and further independent sample-level testing would be beneficial before industrial deployment.

Mean Average Precision (mAP@0.5 and mAP@0.5:0.95) was not used because the developed model performs image classification rather than object detection. Similarly, although the test predictions can be used to generate a confusion matrix, the numerical confusion matrix was not available in the provided training results.

Trained Model Outputs

The model development process produced several output artifacts for subsequent integration and evaluation. The best-performing model checkpoint was saved as black_tea_powder_convnextv2_tiny_best.pth. The class-label mapping was stored in black_tea_powder_class_names.json, while the training history was recorded in black_tea_powder_training_history.csv. The test classification results were stored in black_tea_powder_test_report.txt.

The trained model was subsequently made available through a remotely hosted Hugging Face Space, allowing the TeaGuard AI application to communicate with the classifier through a Gradio-compatible API. The application therefore uses the trained model as an external inference service rather than performing model inference directly within the React frontend or Node.js application.

2.1.5 System architecture

The TeaGuard AI application was implemented using a microservice architecture. This architecture separates major application responsibilities into independently organized services, allowing authentication, request routing, and model integration to be handled by distinct components.

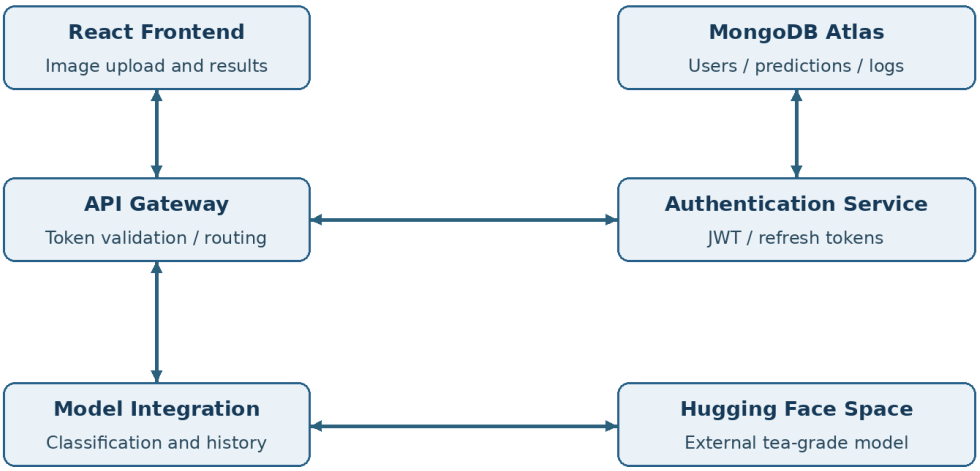


Figure 2:3 Architecture overview

The implemented architecture consists of the following components:

1. Frontend Service

The frontend was developed using React.js, Vite, Tailwind CSS, JavaScript, and React Router. Redux Toolkit and React Redux were used for client-side state management.

The frontend provides the user interface for uploading tea images, initiating classification requests, viewing predicted grades, reviewing previous classification records, and accessing reporting functionality. It communicates with the backend through HTTP requests and REST-style API endpoints.

2. API Gateway

The API Gateway was implemented using Node.js and Express.js and configured to operate on port 5000. It acts as the main entry point for classification requests from the frontend.

The gateway verifies access tokens, validates uploaded image types and file sizes, processes the uploaded image data, and forwards classification requests to the Model

Service. It also separates client-facing API requests from internal service communication.

3. Authentication Service

The Authentication Service was implemented using Node.js and Express.js and configured to operate on port 5001.

It manages user registration, login, token issuance, token refreshing, and related authentication functions. JSON Web Tokens (JWTs) are used to support authenticated access to protected application functionality.

4. Model Service

The Model Service was implemented using Node.js and Express.js and configured to operate on port 5002.

It receives classification requests from the API Gateway, converts the image information into the format required by the external model endpoint, submits inference requests, interprets the returned classification response, and stores prediction records in the database.

5. External Classification Service

The machine learning classifier is hosted remotely through Hugging Face Spaces and accessed using a Gradio-compatible API. The Model Service communicates with this endpoint to obtain tea-grade predictions.

The frontend describes the classification model as ConvNeXt V2 Tiny. However, the exact model architecture, training configuration, and underlying model weights could not be independently verified from the supplied application source code.

6. Database Service

MongoDB Atlas was used as the persistent database, with Mongoose serving as the Object Data Modeling (ODM) library.

The database stores user information and tea-grade classification records. The prediction schema supports information such as the classification module, image metadata, prediction results, the user associated with the record, and timestamps.

Docker and Docker Compose were configured to support containerization of the frontend and backend services.

2.1.6 Tea grade classification categories

The application supports nine processed tea-grade categories. These categories are represented by predefined labels used to interpret and display the model's classification output.

Table 2.1 Tea grade categories

No	Tea grade label
1	BM
2	BP
3	BT
4	BOP
5	BROKEN_TEA
6	DUST
7	FANNING_2
8	PF
9	PW_DUST

The application processes one uploaded image at a time and returns a predicted grade for the complete image. It does not produce individual-particle bounding boxes through the tea-grade classification endpoint.

2.1.7 Image upload and input validation

The classification workflow begins when a user selects or drags and drops a tea sample image into the classification interface.

The frontend submits the selected image to the API Gateway through the endpoint. The image is transmitted with the uploaded file provided through a field named. The request includes a bearer access token for authenticated access.

Multer middleware is used to handle uploaded images in memory. The backend enforces a maximum upload size of 10 MB and checks whether the uploaded file's MIME type begins with image/.

These checks help prevent unsupported file types and excessively large uploads from being passed through the classification workflow. The user interface recommends PNG, JPG, and JPEG image formats.

No explicit image-resizing or normalization operation was identified in the supplied application-level classification workflow. Such preprocessing may occur within the remotely hosted model, but this could not be established from the available source code.

2.1.8 API Gateway Processing

After receiving the uploaded image, the API Gateway performs the required request validation and prepares the image for communication with the Model Service.

The image is initially received as a binary buffer. The gateway converts the buffer into a Base64-encoded representation and forwards the image information to the internal classification endpoint:

POST /internal/models/tea-grade-classify

The internal request is transmitted as JSON and includes the Base64 image data, filename, and MIME type.

This process separates the browser-facing upload mechanism from the format required for communication between the backend services. The gateway also provides a central point for access-token verification and request validation before the classification request reaches the Model Service.

2.1.9 Communication and security

Communication between the frontend and backend is handled through HTTP-based API requests, using multipart form data for image uploads and JSON for backend-to-backend communication.

The main communication interfaces are summarized below.

Table 2.2 API Endpoints and Communication Interfaces of the System

Interface	**Communication purpose**
POST /api/models/tea-grade-classify	Transfers an uploaded image from the frontend to the API Gateway.
POST /internal/models/tea-grade-classify	Transfers encoded image data and metadata from the gateway to the Model Service.
Gradio-compatible API	Submits the image to the remotely hosted classifier and receives inference results.
MongoDB connection	Stores and retrieves user information and prediction records.

JWT-based authentication is used to protect relevant application endpoints. The API Gateway verifies access tokens before processing protected classification requests, while the Authentication Service manages the associated authentication workflow.

Docker and Docker Compose are used to configure the application services within a containerized environment. This supports consistent service configuration and separation of the major application components.

2.1.10 System workflow summary

The complete tea-grade classification workflow begins when the user accesses the TeaGuard AI application and selects a tea sample image for classification. The React frontend sends the selected image to the API Gateway using a multipart/form-data request. Upon receiving the request, the API Gateway verifies the user's access token and validates the uploaded image by checking its MIME type and file size. The image buffer is then converted into Base64 format and forwarded, together with the relevant image metadata, to the Model Service. The Model Service prepares the image in the required Gradio-compatible format and sends it to the remotely hosted tea-grade classification model for inference. The external classification service returns the predicted tea grade along with any available confidence information. The Model Service interprets the returned response and stores the classification record in MongoDB Atlas for future reference. The prediction is then made available to the frontend, where the user can view the classification result and access relevant classification history. In addition, users can review previously stored classification records and export selected information through the application's PDF or CSV reporting functionality. Overall, this workflow demonstrates the interaction between the frontend, backend services, external machine learning classification endpoint, and database in delivering the tea-grade classification functionality.

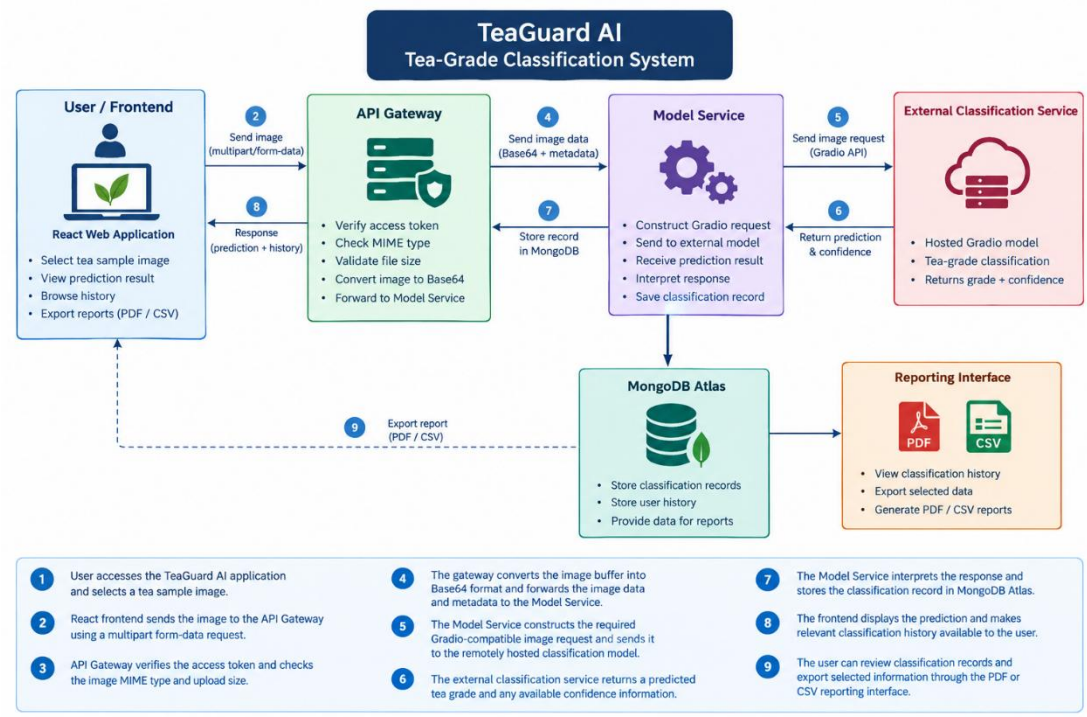


Figure 2:4 System Diagram

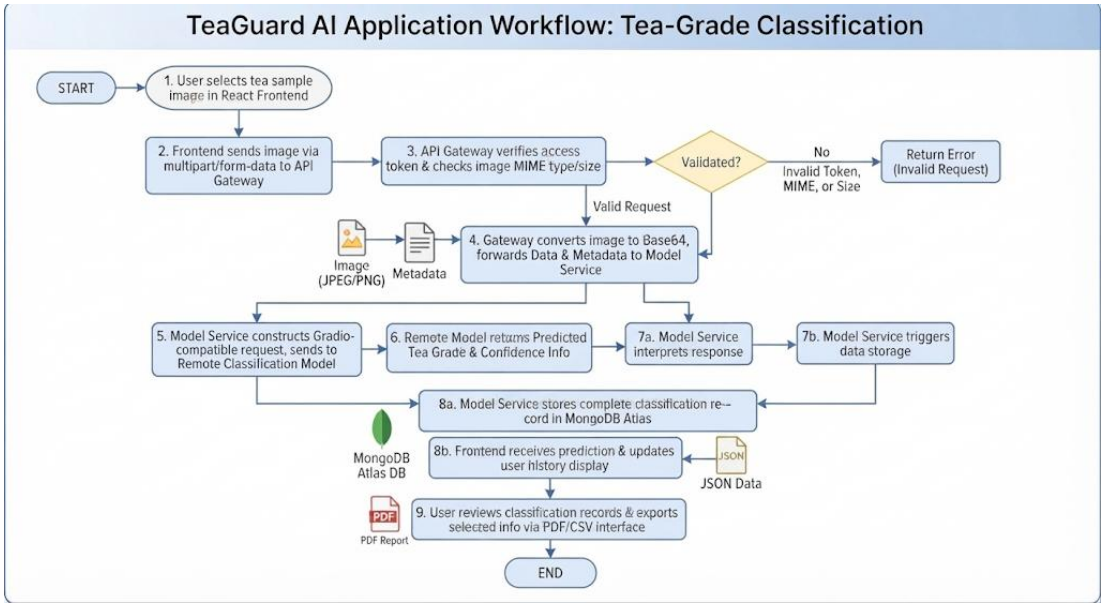


Figure 2:5 Flow chart of the system

2.2 Commercial Aspect of the Product

Commercial Potential

TeaGuard AI has potential to be developed as a commercially viable solution for tea factories, tea-processing companies, quality-control departments, and other organizations involved in processed tea grading. The system provides an automated image-based approach to tea-grade classification, which can support existing quality-control activities by providing consistent classification results through a digital platform. Rather than completely replacing experienced quality-control personnel, the system can be positioned as a decision-support tool that assists them in reviewing and documenting tea-grade classifications.

Product Value

The commercial value of TeaGuard AI is mainly derived from its ability to combine tea-grade classification with digital record management and reporting within a single web-based application. Users can upload tea sample images, obtain a predicted grade, review previous classification records, and export relevant information through PDF and CSV reports. This can reduce dependence on manually maintained classification records and provide a more organized approach to storing and reviewing grading information.

Business and Revenue Model

The proposed product can be offered to tea factories through different commercial models. A factory-based licensing model could provide access to the system for a defined period, such as an annual subscription. Alternatively, the system could be offered as a customized solution based on the requirements of individual tea-processing organizations. Additional services such as system deployment, customization, technical support, maintenance, and integration with existing factory information systems could also provide potential revenue opportunities.

Target Market

The primary target market consists of Sri Lankan tea factories and tea-processing organizations that handle large volumes of processed tea and require regular quality assessment. The system may also have potential applications in tea exporters, quality-assurance organizations, research institutions, and educational or training environments related to tea processing. Since the current system supports nine processed tea-grade categories, future commercial versions could be expanded to include additional grades and organization-specific classification requirements.

Operational and Infrastructure Costs

From a cost perspective, the use of a web-based architecture and remotely hosted machine learning service can provide flexibility when deploying the system. However, commercial deployment would involve operational costs such as cloud hosting, database services, machine learning inference, domain and security requirements, software maintenance, and technical support. Therefore, the final pricing model would need to consider both the organization's usage requirements and the infrastructure costs associated with operating the system.

Future Commercial Opportunities

TeaGuard AI also provides opportunities for future commercialization through additional features. These may include integration with factory production systems, centralized dashboards for multiple production facilities, advanced quality-control analytics, larger-scale batch analysis, additional tea grades, multilingual interfaces, and integration with automated sorting machinery. Such extensions could increase the practical value of the product and support its adoption in larger tea-processing environments.

Overall Commercial Feasibility

Overall, TeaGuard AI demonstrates commercial potential as a digital decision-support solution for processed tea-grade classification. Its combination of automated image-based classification, classification history, and digital reporting provides a foundation for developing the research prototype into a more comprehensive commercial product.

However, large-scale commercial deployment would require further validation using industry-scale datasets, evaluation under real factory operating conditions, assessment of operational costs, and collaboration with tea-processing organizations.

2.3 Consideration of Social, Security and Ethical Aspects

The development of TeaGuard AI considers social, security, and ethical aspects to ensure that the proposed tea-grade classification system can be used responsibly in a practical tea-processing environment. Since the system applies machine learning to classify processed tea samples, attention is given to its potential impact on quality-control practices, the protection of system data, and the responsible use of automated predictions.

Social Aspects

TeaGuard AI has the potential to support the Sri Lankan tea industry by introducing an AI-based approach to processed tea-grade classification. The system can assist tea-factory personnel and quality-control officers by providing image-based classification results and maintaining digital records of previous predictions. This may improve the organization and accessibility of grading information while supporting existing quality-control procedures. The system is intended to assist human decision-making rather than completely replace experienced quality-control personnel. Human verification remains important, particularly when classification results influence production decisions or commercial activities. Furthermore, the system's usefulness in real factory environments depends on its ability to handle variations in tea samples, image-capture conditions, and operational requirements.

Security Aspects

Security is an important consideration because TeaGuard AI includes user authentication, image uploads, backend services, and database storage. The application uses token-based authentication to control access to protected endpoints. Uploaded

images are checked according to the supported MIME types and maximum file-size limit before being processed. Communication between the API Gateway, Model Service, and external classification endpoint must be appropriately protected during deployment. Classification records and associated image metadata stored in MongoDB Atlas should be accessible only to authorized users. In addition, secure configuration of database credentials, authentication secrets, and external service credentials is necessary to reduce the risk of unauthorized access. Production deployment should use HTTPS, appropriate access permissions, secure secret management, and regular software updates to strengthen the overall security of the application.

Ethical Aspects

The ethical implementation of TeaGuard AI requires responsible handling of classification results and transparency regarding the system's capabilities and limitations. Machine learning predictions may not always be correct, and the available confidence information should not be interpreted as a guarantee of classification accuracy. Therefore, users should be encouraged to review predictions and consult qualified quality-control personnel when necessary. The system should not be presented as a complete replacement for established tea-grading standards or professional judgment.

Dataset quality is another important ethical consideration. Tea samples should be labelled using appropriate grading knowledge, and the dataset should represent the intended tea-grade categories as accurately as possible. If the training dataset does not adequately represent variations in sample appearance or image-capture conditions, classification performance may differ when the system is used with new samples. Accordingly, model performance should be evaluated using suitable testing procedures, and limitations should be clearly documented. Data collected from tea factories should also be handled according to any applicable permissions, confidentiality agreements, and organizational requirements. Where factory-sensitive information is involved, it should not be disclosed without authorization.

Overall, TeaGuard AI is designed to provide technological support for processed tea-grade classification while recognizing the importance of human oversight, secure data handling, and responsible use of machine learning. Addressing these social, security, and ethical considerations can help improve the system's trustworthiness and provide a foundation for its future evaluation and potential adoption in the Sri Lankan tea industry.

2.4 Testing and Implementation

2.4.1 System implementation

The system implementation phase translates the proposed system design and technical architecture into a functional web-based application. This section describes the technologies, development tools, and implementation approach used to develop TeaGuard AI, an AI-based tea-grade classification system for processed tea in Sri Lanka. The application integrates a web-based user interface, backend services, a remotely hosted machine learning classification model, and a database to support tea-grade predictions, classification history, and digital reporting.

Technology Stack

Frontend Technologies

- **React.js** – A JavaScript library used to develop the user interface through reusable components. It enables users to upload tea sample images, view classification results, access previous classification records, and use the reporting functionality.
- **JavaScript** – Used to implement frontend logic, handle user interactions, manage application functionality, and communicate with backend APIs.
- **Vite** – A frontend development and build tool used to support application development and prepare the frontend for deployment.
- **Tailwind CSS** – A utility-first CSS framework used to design a consistent, responsive, and user-friendly interface.
- **React Router** – Used to manage navigation between application pages.
- **Redux Toolkit and React Redux** – Used to manage shared application state where required.

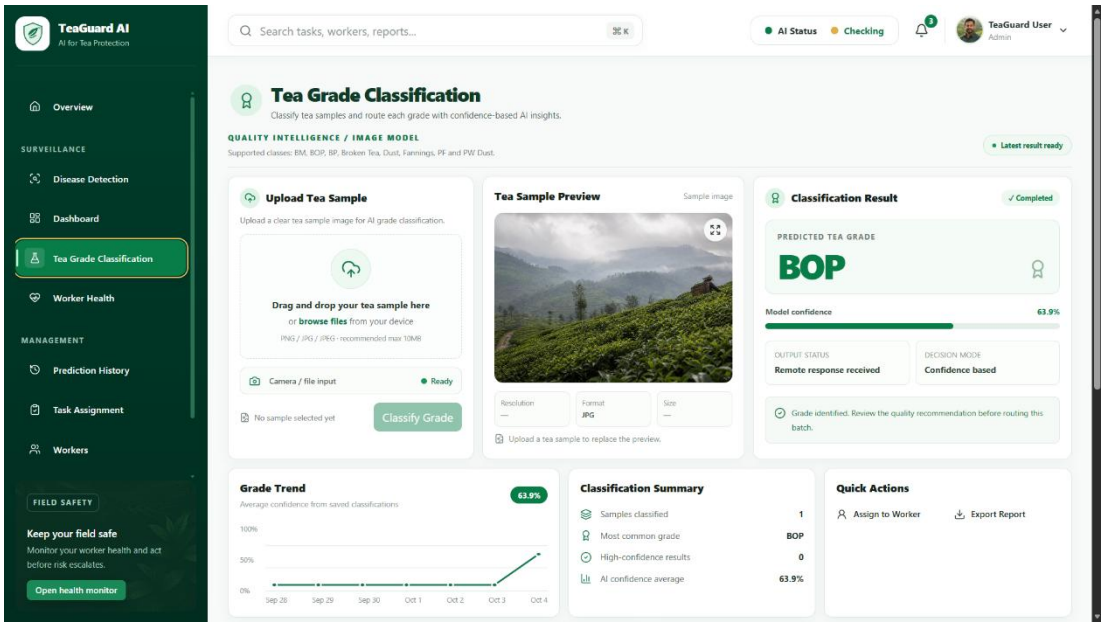


Figure 2:6 Sample UI of a system

Backend Technologies

- **Node.js** – A JavaScript runtime environment used to implement the backend services that handle application requests and business logic.
- **Express.js** – A web framework used to develop the API Gateway, authentication service, and Model Service. These services manage client requests, authentication, image processing, communication with the classification endpoint, and the delivery of prediction results.
- **Multer** – A middleware library used to process uploaded images through multipart/form-data requests. The application validates image MIME types and applies an upload size limit before forwarding the image for classification.
- **JSON Web Tokens (JWT)** – Used to support token-based authentication and authorization for protected application endpoints.

Machine Learning and Image Classification

- **Hosted Machine Learning Classification Model** – TeaGuard AI integrates a remotely hosted image-classification model to predict the grade of an uploaded processed tea sample. The Model Service communicates with the external

classification endpoint through a Gradio-compatible interface and processes the returned prediction.

- **Gradio-Compatible API Integration** – Used to send image data to the hosted classification service and receive the classification response.
- **Base64 Encoding** – Used by the API Gateway to convert the uploaded image buffer into a Base64 representation before forwarding the image data and associated metadata to the Model Service.

The system supports classification into nine processed tea-grade categories: BM, BOP, BT, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust. The predicted grade and any available confidence information are processed and returned to the frontend for display. The exact model architecture, training configuration, and evaluation metrics should be documented separately using the actual model-development records.

Database Technologies

- **MongoDB Atlas** – A cloud-hosted NoSQL database service used to store classification records and relevant image metadata. It supports the retrieval of historical predictions and provides persistent storage for information required by the application's history and reporting functions.
- **Mongoose** – An Object Data Modeling (ODM) library used by the Model Service to define data schemas and interact with MongoDB Atlas.

Reporting and Data Management

TeaGuard AI includes classification history and digital reporting functionality. Users can review stored classification records and export selected information in PDF or CSV format. These reports are generated through the application's reporting interface, supporting the organized presentation and management of classification information.

Development and Design Tools

- **Visual Studio Code (VS Code)** – Used as the development environment for writing, editing, debugging, and maintaining the frontend and backend source code.
- **Git and GitHub**, if used in the project – Used for source-code version control, repository management, and tracking development changes.

Deployment and Development Infrastructure

- **Docker** – Used to package application services into containers to support consistent execution across development and deployment environments.
- **Docker Compose** – Used to configure and coordinate the application's containerized services.
- **Hugging Face Spaces** – Used to host the external machine learning classification endpoint accessed by the Model Service.

Testing and Collaboration Tools

- **Postman** – An API development and testing tool used to test backend endpoints, validate request and response formats, and verify communication between the frontend, API Gateway, authentication service, and Model Service.
- **Microsoft Teams and WhatsApp** – Communication and collaboration platforms used to facilitate team coordination, discussions, meetings, and the exchange of project-related information during system development.
- **Google Drive** – A cloud-based storage platform used to organize and share research documents, technical documentation, design assets, and other project deliverables among team members.

The application follows a service-based architecture consisting of the React frontend, API Gateway, authentication service, Model Service, MongoDB Atlas database, and remotely hosted classification model. The API Gateway operates on port 5000, the authentication service on port 5001, and the Model Service on port 5002 in the

documented development configuration. This separation of responsibilities supports maintainability and allows individual services to be managed independently.

Overall, the selected technologies provide the foundation for implementing TeaGuard AI as a web-based tea-grade classification platform. The integration of image-based machine learning, backend services, database storage, and digital reporting enables the system to deliver classification functionality while maintaining a structure that can be extended and improved in future development.

2.4.2 System testing

Testing Methodologies

The TeaGuard AI system was evaluated through testing activities designed to verify its functionality, integration, reliability, and usability. The testing focused on the main system features, including user authentication, tea sample image uploading, tea-grade classification, database operations, classification history, and digital report generation.

- **Unit Testing:** Individual frontend components and backend functions were checked to verify that they performed their intended tasks. Testing focused on image validation, user input handling, authentication-related functions, and the processing of classification results.
- **Integration Testing:** Integration testing was used to verify communication between the React frontend, API Gateway, authentication service, Model Service, external machine learning classification endpoint, and MongoDB Atlas database. API requests and responses were tested using Postman. The testing also covered the storage and retrieval of classification records and the generation of prediction results.
- **Functional Testing:** The main system functions were tested using predefined test cases. These included user logins, uploading valid tea sample images, rejecting unsupported file formats, obtaining predicted tea grades, viewing classification history, and exporting reports in PDF and CSV formats. The

actual results were compared with the expected results to identify functional errors.

- **Machine Learning Model Testing:** The classification model was evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The model achieved a test accuracy of 94.26% and a macro F1-score of 94.72% across the nine processed tea-grade categories. These metrics were used to assess classification performance and identify grades that were more difficult to distinguish.
- **Performance Testing:** The system's response time and behavior during image uploads and classification requests are relevant to evaluating its suitability for practical use. Load testing with multiple concurrent users would help determine how well the application handles increased demand. Quantitative load-testing results should be included only if these tests were performed, and measurements were recorded.
- **Cross-Browser Testing:** The web application can be checked using commonly used browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge, to verify that the interface, image-upload process, classification results, navigation, and reporting features work consistently. Browser compatibility results should be documented according to the browsers actually tested.

Overall, these testing methodologies provide a structured approach to evaluating the TeaGuard AI system. They help verify the correct operation of the application's main features, the communication between its components, and the classification model's performance. Additional performance and browser compatibility testing can further establish the system's readiness for use in practical tea-processing environments.

Test Cases Design

System testing was conducted to evaluate the functionality, reliability, and usability of the TeaGuard AI web-based tea-grade classification system. The testing focused on verifying that the frontend, backend services, externally hosted machine learning model, and MongoDB Atlas database work together as expected. Functional test cases

were prepared to assess user authentication, image upload and validation, tea-grade classification, prediction history, and report generation. Each test case was executed using predefined inputs and expected outcomes to identify functional errors and verify that the implemented system satisfies its requirements.

Table 2.3: TG_TC_01

Field	Details
Test Case ID	TG_TC_01
Test Case Scenario	Upload Valid Tea Sample Image
Description	To verify that the system accepts a valid tea sample image for classification.
Preconditions	The user is logged in and the tea-grade classification page is accessible.
Test Steps	1. Select the image upload option. 2. Select a valid tea sample image in PNG, JPG, or JPEG format. 3. Submit the image for classification. 4. Observe the upload process.
Expected Results	The system accepts the image and proceeds with the tea-grade classification process.
Actual Results	To be recorded during testing
Status	Pass

Table 2.4: TG_TC_02

Field	Details
Test Case ID	TG_TC_02
Test Case Scenario	Validation of Unsupported Image File
Description	To verify that the system rejects an uploaded file that does not meet the supported image requirements.
Preconditions	The user is logged in and the classification page is accessible.
Test Steps	1. Select the image upload option. 2. Attempt to upload an unsupported file type. 3. Observe the system response.
Expected Results	The system rejects the unsupported file and displays an appropriate validation message.

Actual Results	To be recorded during testing
Status	Pass

Table 2.5: TG_TC_03

Field	**Details**
Test Case ID	TG_TC_03
Test Case Scenario	Tea Grade Classification
Description	To verify that the system successfully sends a valid tea sample image to the machine learning model and displays the predicted tea grade.
Preconditions	The user is logged in and a valid tea sample image is available.
Test Steps	1. Upload a valid tea sample image. 2. Submit the image for classification. 3. Wait for the classification response. 4. Observe the predicted result.
Expected Results	The system successfully processes the image and displays the predicted tea grade returned by the machine learning model.
Actual Results	To be recorded during testing
Status	Pass

Table 2.6: TG_TC_04

Field	**Details**
Test Case ID	TG_TC_04
Test Case Scenario	View Classification History
Description	To verify that users can access and view their previously stored tea-grade classification records.
Preconditions	The user is logged in and previous classification records are available.
Test Steps	1. Navigate to the classification history section. 2. View the available records. 3. Observe the displayed results.
Expected Results	The system displays the user's stored classification records correctly and applies the available search or filtering options.

Actual Results	To be recorded during testing
Status	Pass

Table 2.7: TG_TC_05

Field	Details
Test Case ID	TG_TC_05
Test Case Scenario	Export Classification Data as CSV
Description	To verify that classification records can be exported in CSV format.
Preconditions	The user is logged in and classification records are available.
Test Steps	1. Open the reporting section. 2. Select the required records or filters. 3. Select the CSV export option. 4. Open the generated CSV file.
Expected Results	The system successfully generates a CSV file containing the selected classification data.
Actual Results	To be recorded during testing
Status	Pass

3. RESULTS AND DISCUSSION

3.1 Results

The results show that the proposed TeaGuard AI system achieved good performance in classifying processed tea grades using tea sample images. A total of 1,217 usable images were collected across nine tea-grade categories: BM, BP, BT, BOP, Broken Tea, Dust, Fanning 2, PF, and PW Dust. The dataset was divided into training, validation, and testing sets using an 80:10:10 ratio, resulting in 973 training images, 122 validation images, and 122 testing images. The ConvNeXt V2 Tiny model was trained to identify the different tea grades based on the input images.

During training, the model achieved a training accuracy of 98.66% and a training F1-score of 98.62%. The validation accuracy reached 95.08%, while the validation F1-score was 95.60%. The best validation macro F1-score was approximately 96.44%, indicating that the model performed well across the different tea-grade categories during training and validation.

When evaluated using the test dataset, the model achieved an accuracy of 94.26% and a macro F1-score of 94.72%. Out of the 122 test images, approximately 115 were classified correctly. These results indicate that the model was able to identify most of the tea grades correctly. The test results also show that the model maintained strong performance when evaluated using images that were not included in the training process.

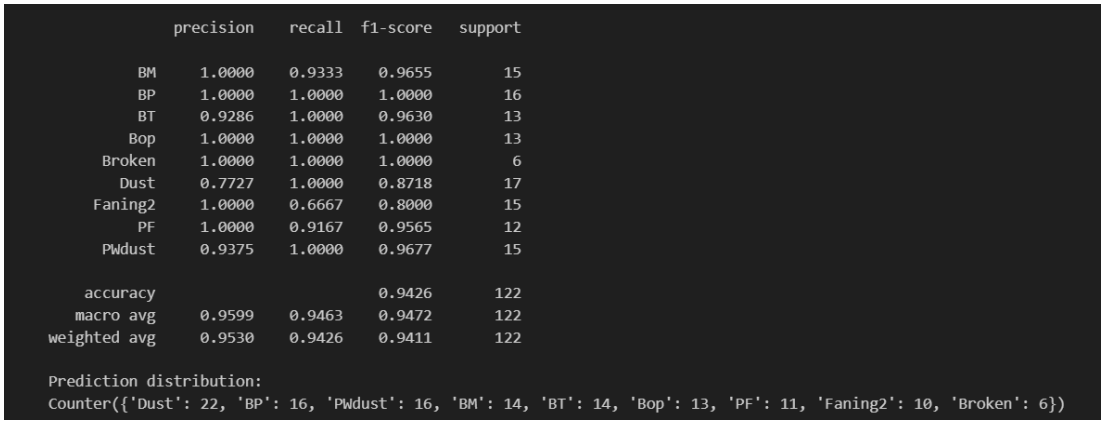
The class-level results showed that BP, BOP, and Broken Tea achieved precision, recall, and F1-scores of 100%. However, the Broken Tea result was based on only six test images, so further testing with a larger number of samples is needed to confirm its performance. BT achieved an F1-score of 96.30%, while PW Dust achieved 96.77%. BM and PF also showed strong results, with F1-scores of 96.55% and 95.65%, respectively.

Some differences were observed in the classification performance of Dust and Fanning 2. Dust achieved a recall of 100%, meaning that all actual Dust samples in the test set

were identified. However, its precision was 77.27%, indicating that some images belonging to other tea grades were incorrectly classified as Dust. Fanning 2 achieved a precision of 100% but a recall of 66.67%. This means that although the images predicted as Fanning 2 were correct, some actual Fanning 2 samples were classified as other grades. These results suggest that distinguishing between certain tea grades remains a challenge for the model.

The accuracy and loss curves provide further information about the model's training process, while the macro F1-score curve shows its performance across the different classes. The confusion matrix also helps identify which tea grades were correctly classified and where classification errors occurred. Together, these results provide a clearer understanding of the model's strengths and limitations.

Overall, the experimental results demonstrate that TeaGuard AI can classify the nine processed tea grades with a test accuracy of 94.26%. The findings suggest that the system has potential to support tea-grade identification and reduce dependence on manual inspection. However, additional testing using images from different tea batches, lighting conditions, and real processing environments is necessary before the system can be considered reliable for practical use in tea factories.



	precision	recall	f1-score	support
BM	1.0000	0.9333	0.9655	15
BP	1.0000	1.0000	1.0000	16
BT	0.9286	1.0000	0.9630	13
Bop	1.0000	1.0000	1.0000	13
Broken	1.0000	1.0000	1.0000	6
Dust	0.7727	1.0000	0.8718	17
Faning2	1.0000	0.6667	0.8000	15
PF	1.0000	0.9167	0.9565	12
PWdust	0.9375	1.0000	0.9677	15
accuracy			0.9426	122
macro avg	0.9599	0.9463	0.9472	122
weighted avg	0.9530	0.9426	0.9411	122

Prediction distribution:
Counter({'Dust': 22, 'BP': 16, 'PWdust': 16, 'BM': 14, 'BT': 14, 'Bop': 13, 'PF': 11, 'Faning2': 10, 'Broken': 6})

Figure 3:1 Model Accuracy

The accuracy curve shows that the model's training accuracy increased steadily during the training process and reached approximately 98–99%. The validation accuracy also improved and remained around 95% towards the end of training. This indicates that

the model learned to classify the different tea grades effectively. The small difference between the training and validation accuracy suggests that the model maintained good performance on the validation dataset.

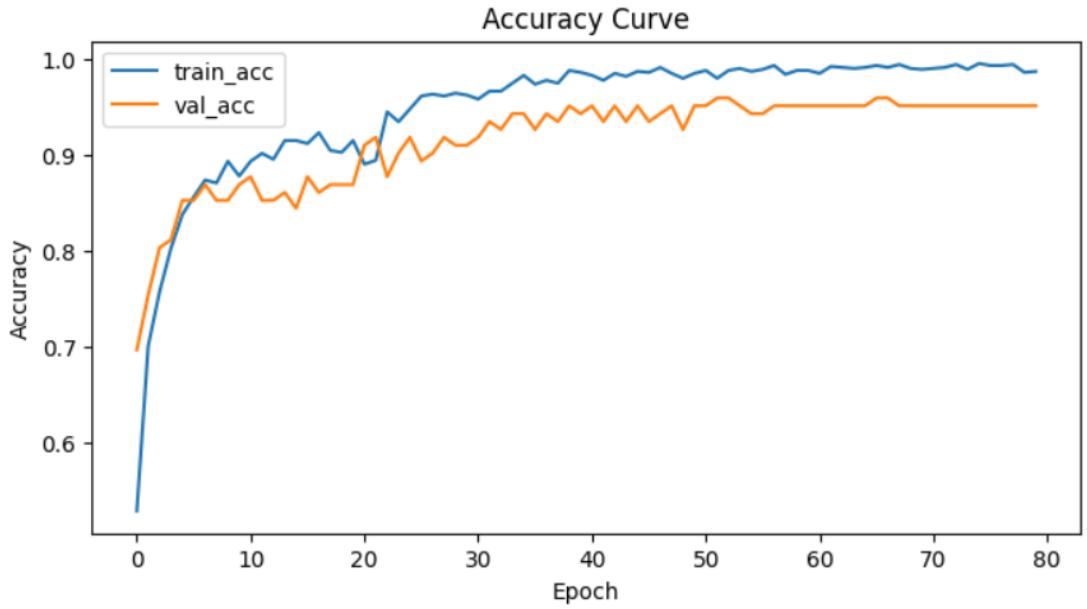


Figure 3:2 accuracy curve

The macro F1 curve shows a similar improvement throughout training. The training macro F1-score reached approximately 98–99%, while the validation macro F1-score remained around 95–96%. This indicates that the model performed well across the different tea-grade categories. The relatively stable validation results during the later epochs suggest that the model's performance had largely stabilized.

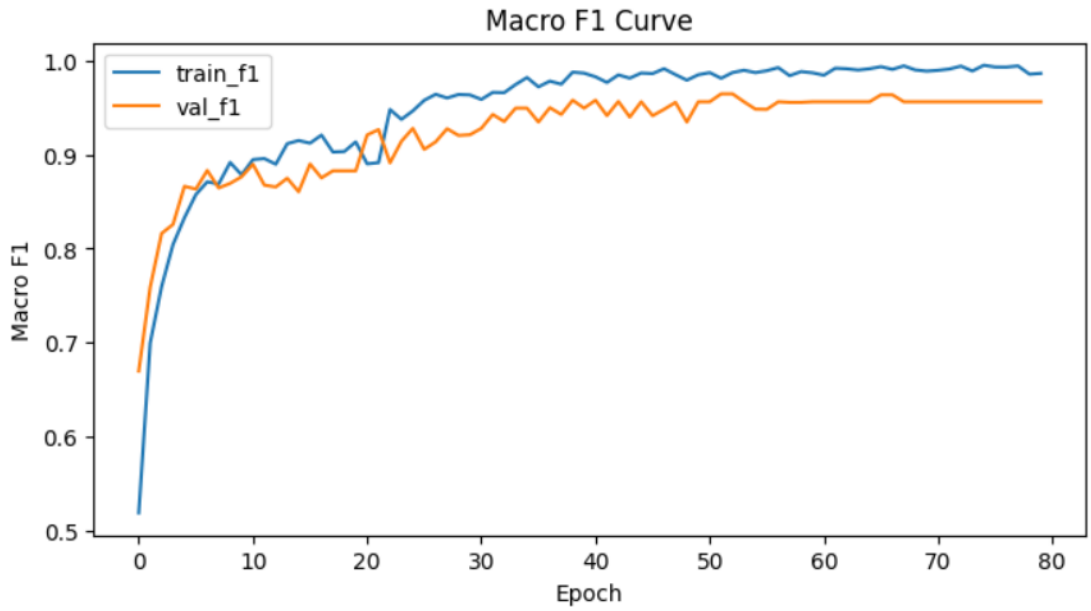


Figure 3:3 Macro F1 Curve

The confusion matrix provides a detailed view of the model's classification results on the test dataset. The model correctly classified 115 out of 122 images. Most tea grades were classified correctly, including BP, BT, BOP, Broken Tea, and Dust. However, five Fanning 2 images were incorrectly classified as Dust, while one BM image was classified as BT and one PF image was classified as PW Dust. These errors show that some tea-grade categories may have visual similarities that make them more difficult to distinguish. Overall, the confusion matrix supports the model's strong classification performance while identifying areas that could be improved in future development.

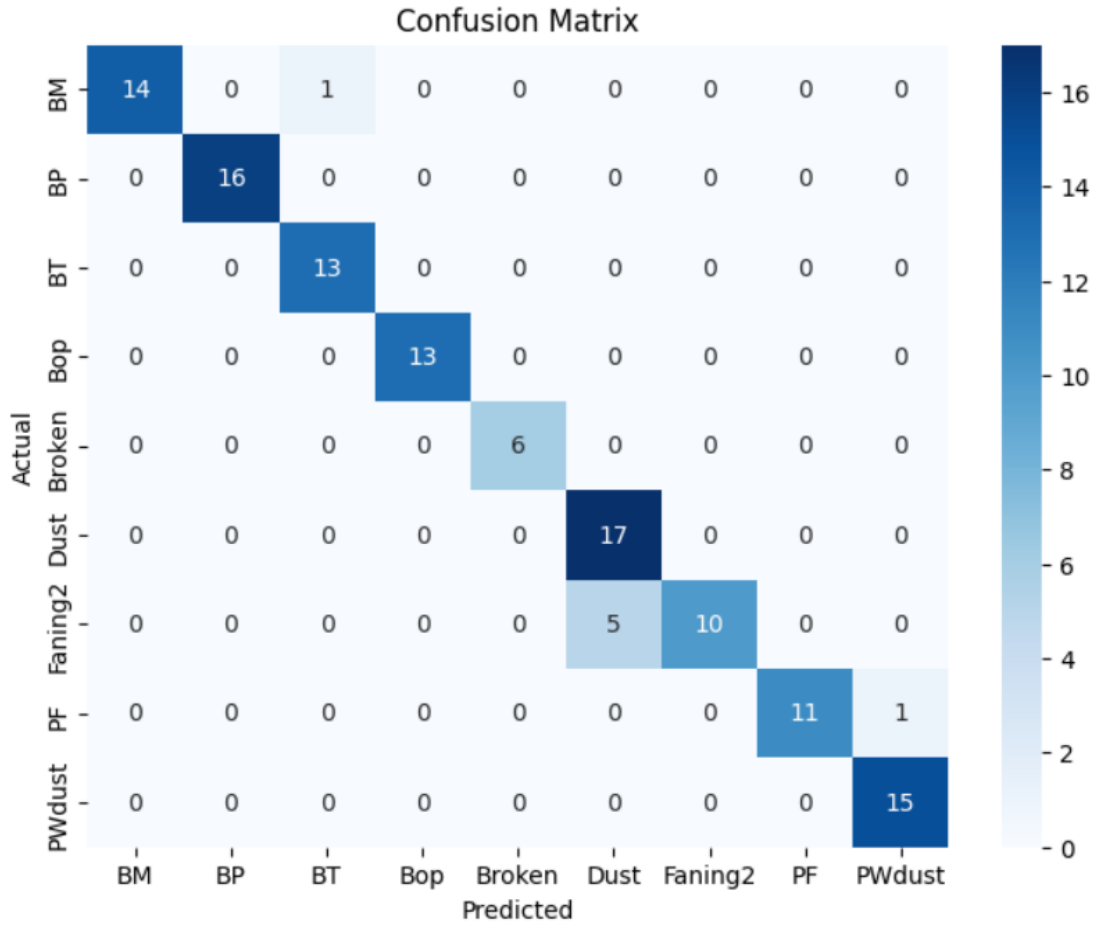


Figure 3:4 Confusion matrix

3.2 Research Findings

The findings of this research demonstrate that an image-based deep learning approach can be used to classify nine processed tea grades relevant to the Sri Lankan tea industry. The developed TeaGuard AI model was evaluated using 1,217 tea sample images representing BM, BP, BT, BOP, Broken Tea, Dust, Fanning 2, PF, and PW Dust. The experimental results showed a test accuracy of 94.26% and a macro F1-score of 94.72%. These findings indicate that the model was able to identify most of the tea grades correctly based on the visual characteristics of the sample images.

The training and validation results showed that the model learned to recognize patterns within the tea images and maintained good classification performance on data not used

for training. The confusion matrix further showed that several tea grades were classified correctly in all their test samples. However, some errors occurred when distinguishing Fanning 2 from Dust, as well as between BM and BT and between PF and PW Dust. This suggests that certain tea grades may share visual characteristics that make accurate classification more challenging.

Another important finding is that the model's performance varied across the nine tea-grade categories. Although the overall accuracy was high, the lower recall for Fanning 2 and lower precision for Dust indicate that further improvements may be needed to distinguish these categories more reliably. Therefore, overall accuracy alone is not sufficient to evaluate the model; examining individual class results is also important.

The research also demonstrates the potential of integrating image-based tea-grade classification into a web-based application. TeaGuard AI provides a platform through which users can upload tea sample images, obtain predicted grades, review previous classification records, and generate digital reports. These features could support the organization of grading information and assist quality-control personnel in their daily activities. However, the practical impact of these functions on grading time, cost, and consistency requires further evaluation in a real tea-factory environment.

Overall, the findings support the potential of using deep learning to assist the classification of processed tea grades. The achieved test results provide a promising foundation for further development of TeaGuard AI as a decision-support system for the Sri Lankan tea industry. Nevertheless, testing with larger datasets collected from different tea batches, factories, and imaging conditions is necessary to establish the reliability and generalizability of the system before practical industrial deployment.

3.3 Discussion

The results of this research demonstrate that the developed TeaGuard AI model has the potential to classify nine processed tea grades using digital images. The model achieved a test accuracy of 94.26% and a macro F1-score of 94.72%, indicating that it

was able to identify most of the tea grades correctly. These results suggest that image-based classification can be a useful approach for supporting tea-grade identification in the Sri Lankan tea industry. The model's performance is encouraging because it was evaluated using a separate test dataset that was not used during training.

The accuracy and macro F1 curves show that the model's performance improved during training and remained relatively stable towards the final epochs. The training accuracy reached approximately 98–99%, while the validation accuracy remained around 95%. A similar pattern was observed in the macro F1 curves. The difference between the training and validation results suggests that the model learned the visual features of the tea samples effectively, although some degree of overfitting may still be present. Further evaluation using completely independent tea samples would help determine how well the model performs on data collected under different conditions.

The confusion matrix provides further insight into the strengths and limitations of the classification model. BP, BT, BOP, Broken Tea, and Dust were correctly identified for all samples in the test set. However, five actual Fanning 2 samples were classified as Dust. This indicates that the model may have difficulty distinguishing these two categories based on their visual appearance. In addition, one BM sample was classified as BT, and one PF sample was classified as PW Dust. These errors demonstrate that, although the overall performance is high, some tea grades may have similar visual characteristics that affect classification accuracy.

The class-level evaluation also highlights the importance of considering precision and recall alongside overall accuracy. Dust achieved 100% recall but a precision of 77.27%, meaning that all actual Dust samples were identified, but some samples from other grades were also incorrectly predicted as Dust. In contrast, Fanning 2 achieved 100% precision but a recall of 66.67%, indicating that some actual Fanning 2 samples were assigned to another category. These findings suggest that improvements should focus particularly on distinguishing visually similar grades and increasing the model's ability to identify all samples belonging to each category. However, the results for classes with relatively few test images should be interpreted carefully, as their performance estimates may change when evaluated using a larger dataset.

The findings also support the potential value of integrating the classification model into the TeaGuard AI web application. By allowing users to upload tea sample images and receive predicted grades, the system provides a digital alternative to performing every classification task through manual visual inspection. Its classification history and reporting features can also help organize grading information and make previous results easier to review. Nevertheless, the present classification metrics do not directly demonstrate reductions in grading time, operating costs, or human error. These benefits would need to be measured through further testing with tea-factory personnel in practical working environments.

Several limitations should be considered when interpreting the results. The evaluation was conducted using 122 test images from the available dataset, which may not represent all the variations found in commercial tea production. Differences in lighting, camera quality, sample arrangement, moisture, and tea batches could affect the model's predictions. In addition, the current results do not establish whether the model will maintain the same accuracy when applied to samples from factories that were not represented in the dataset. Future work should therefore include a larger and more diverse dataset, testing with independent tea batches and factories, and further investigation of classification errors between Dust and Fanning 2.

Overall, the discussion indicates that TeaGuard AI provides a promising foundation for automated processed tea-grade classification. The test accuracy of 94.26% demonstrates encouraging performance across the nine selected grades, while the confusion matrix identifies specific areas requiring improvement. The system should therefore be considered a decision-support tool that can assist quality-control personnel rather than a complete replacement for human expertise. Further real-world validation is necessary to establish its reliability and suitability for use in tea-processing environments.

4. CONCLUSIONS

The TeaGuard AI system was developed to support the classification of processed tea grades using image-based deep learning. The developed model was trained using 1,217 tea sample images representing nine processed tea-grade categories: BM, BP, BT, BOP, Broken Tea, Dust, Fanning 2, PF, and PW Dust. The ConvNeXt V2 Tiny model achieved a test accuracy of 94.26% and a macro F1-score of 94.72%, demonstrating strong performance in identifying the selected tea grades from sample images.

The findings indicate that image-based classification can provide useful technological support for tea-grade identification. The integration of the trained classification model into the TeaGuard AI web application allows users to upload tea sample images, obtain predicted grades, review previous classification records, and generate digital reports. These features provide a structured digital approach for recording and reviewing tea-grade classification results and can assist quality-control personnel in their activities.

However, the results also identified challenges in distinguishing visually similar categories, particularly Fanning 2 and Dust. The classification performance was also evaluated using a relatively limited test dataset images, meaning that the reported performance may not fully represent the variations present in real tea-processing environments. Therefore, the current system should be considered a decision-support solution rather than a complete replacement for human expertise.

Overall, the research demonstrates the potential of deep learning and web-based technologies to support processed tea-grade classification in the Sri Lankan tea industry. Future improvements should focus on expanding the dataset, including samples from different tea batches and factories, evaluating the system under different imaging and environmental conditions, and further investigating classification errors between visually similar tea grades. Such improvements can strengthen the reliability, generalizability, and practical applicability of TeaGuard AI in real-world tea-processing environments.

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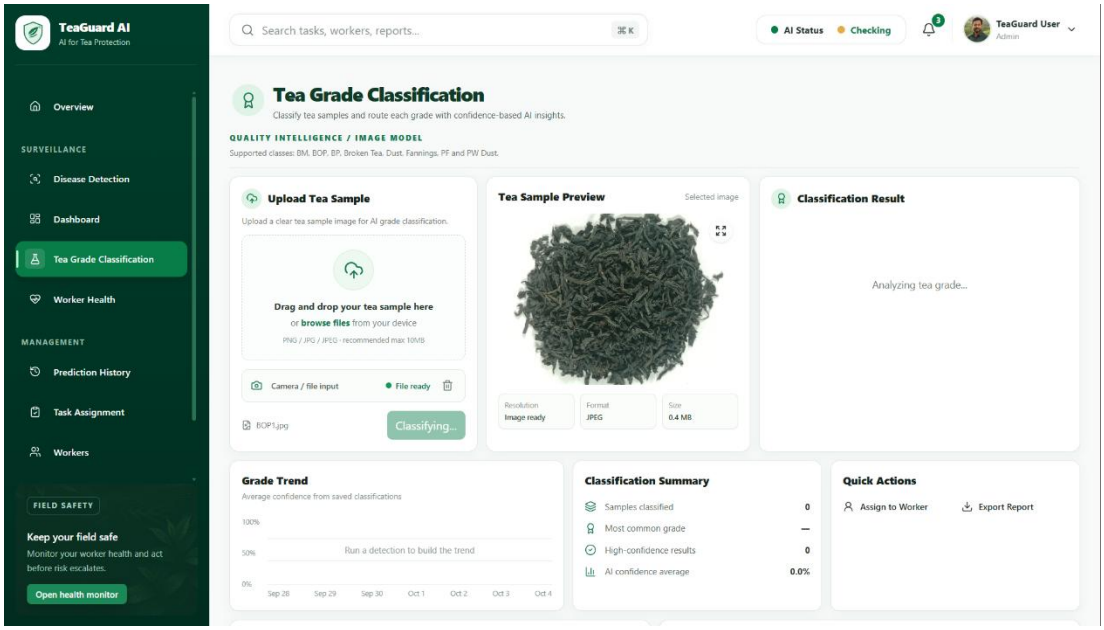
6. GLOSSARY

Term	Definition
Accuracy	A metric that measures the percentage of correctly predicted outcomes.
API (Application Programming Interface)	A set of rules and interfaces that allows different software components to communicate and exchange data.
API Gateway	A backend service that receives client requests and routes them to the appropriate backend service.
Artificial Intelligence (AI)	The field of computing concerned with developing systems that can perform tasks that normally require human intelligence.
CNN (Convolutional Neural Network)	A type of deep learning neural network commonly used for analyzing and classifying image data.
Confidence Score	A value returned by a classification model indicating the confidence associated with a predicted class.
Confusion Matrix	A table used to evaluate a classification model by showing correct and incorrect predictions for each class.
ConvNeXt V2	A modern convolutional neural network architecture used for image classification and visual recognition tasks.
Deep Learning	A branch of machine learning that uses multi-layer neural networks to learn complex patterns from data.
F1-Score	The harmonic means of precision and recall, used for balanced evaluation.
Fanning 2	A processed tea grade included among the tea categories classified by the TeaGuard AI model.
Fine-Tuning	The process of further training a pretrained machine learning model on a specific dataset and task.
Hugging Face	A platform that provides tools and services for developing, hosting, and deploying machine learning models and applications.
Image Classification	The process of assigning an input image to one of a predefined set of classes.
Inference	The process of using a trained machine learning model to generate a prediction for new input data.
Machine Learning (ML)	A field of artificial intelligence where systems learn patterns from data and make predictions without explicit programming.
Macro F1-Score	The average F1-score calculated across all classes, giving equal importance to each class.
MongoDB Atlas	A cloud-based database service used to store and manage application data.

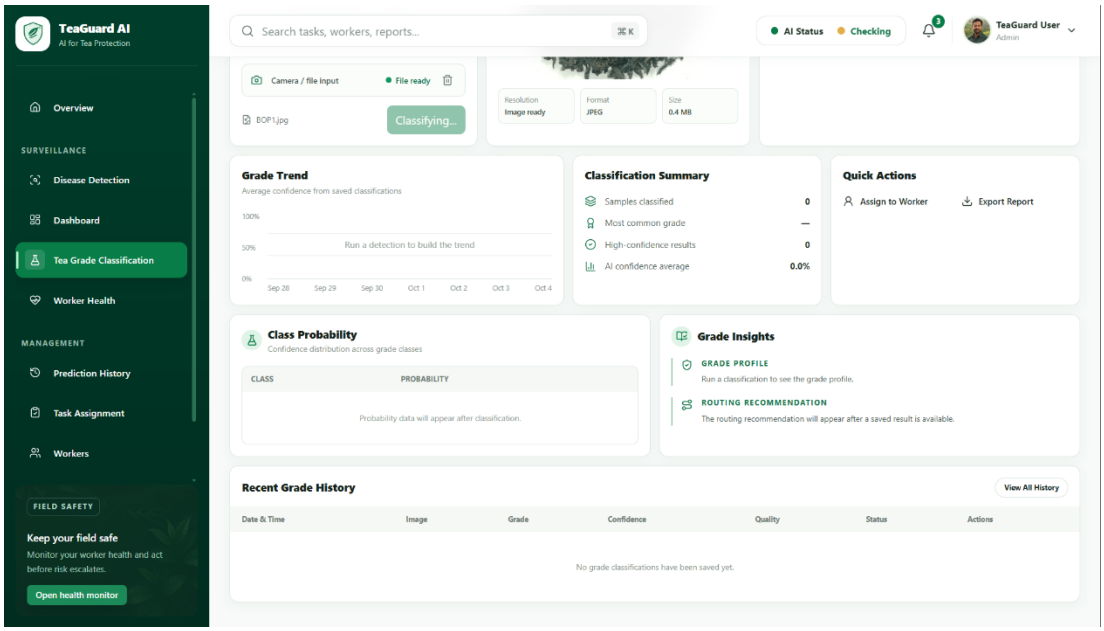
Model Service	A backend service responsible for communicating with the hosted machine learning model and processing classification requests.
Precision	A classification metric that measures the proportion of predicted positive instances that are actually correct.
Recall	A classification metric that measures the proportion of actual positive instances that are correctly identified by the model.
REST API	An application programming interface that enables systems to communicate through HTTP-based requests and responses.
Tea Grade Classification	The process of identifying the appropriate processed tea grade from an input tea sample image.
Transfer Learning	A machine learning approach in which knowledge learned from a pretrained model is adapted to a new but related task.
Validation Dataset	A portion of the dataset used during model development to evaluate performance and monitor the training process.
Test Dataset	A separate portion of the dataset used to evaluate the final performance of a trained machine learning model.
Training Dataset	The portion of the dataset used to train a machine learning model to learn patterns and features.
JWT (JSON Web Token)	A token-based authentication mechanism used to securely transmit information and authenticate users between applications.
Multer	A Node.js middleware used to handle multipart/form-data, particularly file uploads.
Gradio	A framework used to create interfaces for machine learning models and provide model inference through an API.
MongoDB	A NoSQL database that stores data in flexible, document-oriented structures.
Microservice	An independently developed and deployable software service that performs a specific application function.
Web-Based System	A software application is accessed through a web browser without installation.

7. APPENDICES

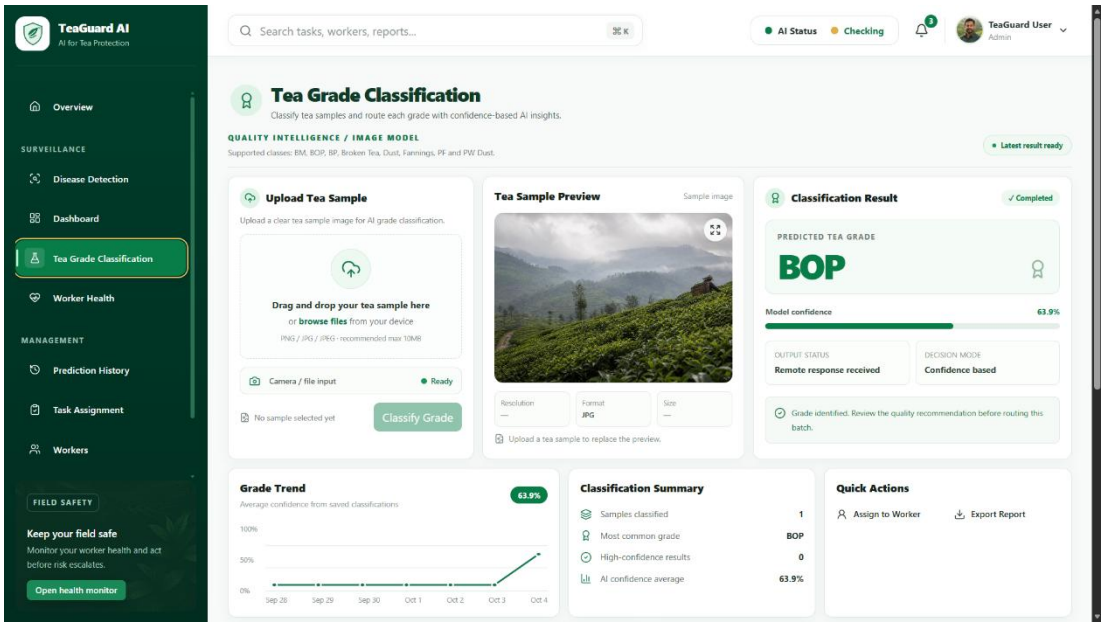
Appendix A: User Interfaces



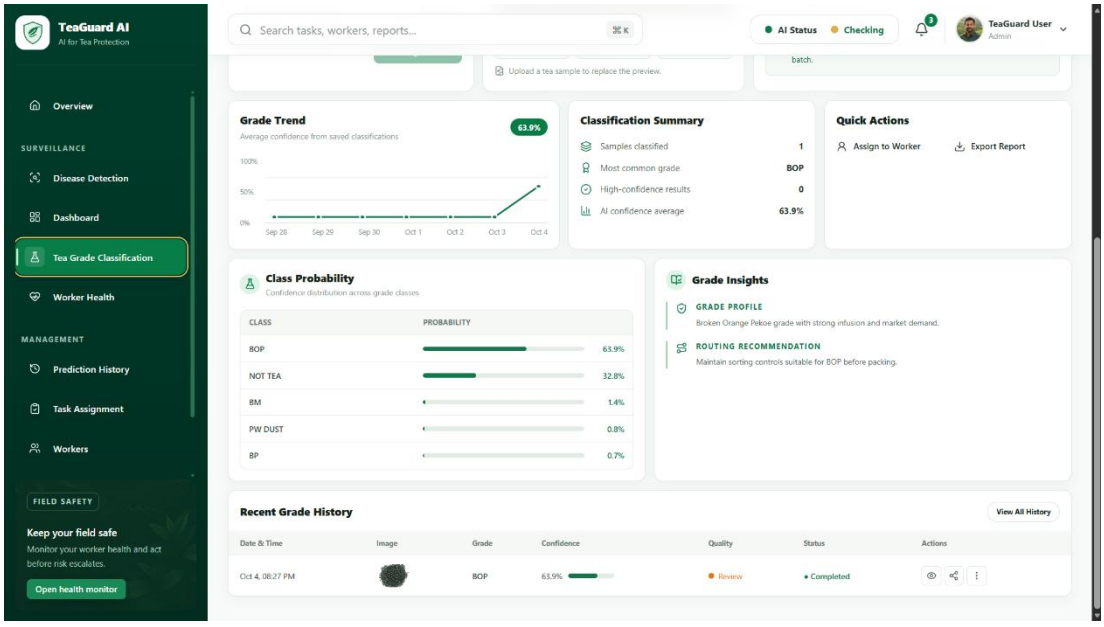
Appendix 1: Sample UI 1



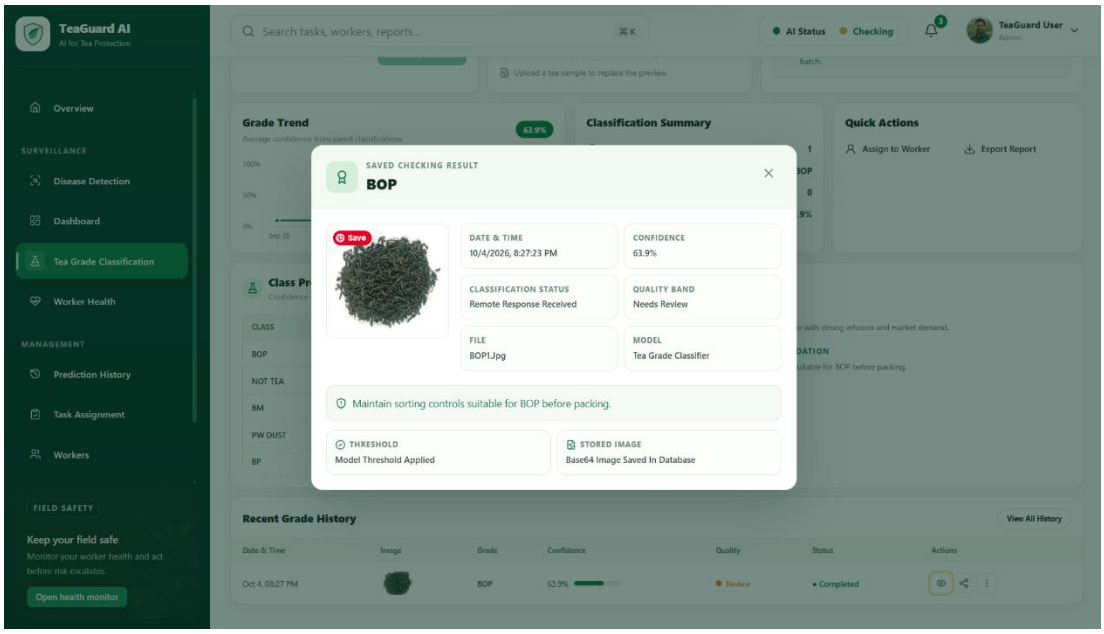
Appendix 2: Sample UI 2



Appendix 3 sample UI 3



Appendix 4 sample UI 4



Appendix 5 sample UI 5